

budget 0.2343 — completes

budget 0.235 — out of memory

Op 2,266
(produces storage 3131)

Same 14 evictions as 0.235, then still
173,479 B short → evicts storage 932
(218,275,840 B)

Same 14 evictions; budget is 7,871,746 B
larger, so no shortfall remains
→ keeps storage 932



Op 2,267
(requests 60,211,200 B)

Fits in the space freed by 932
→ no eviction; 3131 stays resident

49,367,205 B short → evicts lowest h DTR:
3131 ($h = 2.82e-9$), ranked ahead of
932 ($h = 8.18e-9$) among 675 candidates



Op 2,270
(needs storage 3131)

3131 resident → run completes
(1.471× overhead)

3131 evicted → rebuild cascade:
max nesting depth 193; peak pinned 2.63 GB
(depth at peak pinned 188) → out of memory

Intervention at 0.235, same budget: never evicting storage 3131 → completes (1.477×).
Control, same wrapper with no retained tensor → out of memory (peak pinned 2.63 GB; max nesting depth 193; depth at peak pinned 188).

Which tensor is needed later is hindsight from the trace; DTR's score ($e^*/(\text{size} \cdot \text{staleness})$) does not use future accesses. simrd reference simulator; no allocator model.